Word Embeddings - Theory and Analysis

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* Introduction:

A computer cannot understand the word cat. It can only do arithmetic. So every word must first be turned into numbers.

A word embedding is exactly that - a word represent as a list of numbers, called a vector:

"cat" → [0.21, -0.44, 0.87, ...]

What matters is which numbers. Chosen well, words with similar meanings get similar numbers, so a computer can tell that cat is closer to dog than to rocket - without being told what any of them mean.

1. One-Hot Encoding vs. Dense Embeddings

→ One-hot encoding: Each word becomes a vector as long as the whole vocabulary: all zeros, with a single 1 marking its position.

Vocabulary = [cat, dog, king, queen, rocket]

cat = [1, 0, 0, 0, 0]
dog = [0, 1, 0, 0, 0]
king = [0, 0, 1, 0, 0]

This has two serious problems:

(a) Size: A real vocabulary holds about 50,000 words, so every word becomes 50,000 numbers of which 49,999 are zero. Nearly all storage is wasted.

(b) No meaning: Every pair of different words is equally far apart:

$\cos(\text{cat}, \text{dog}) = 0 \qquad \cos(\text{cat}, \text{rocket}) = 0$

The encoding claims cat is as unrelated to dog as it is to rocket.

⇒ Dense embeddings: fix both problems. Each word becomes a short vector of real numbers learned from text. Because cat and dog appear in similar sentences, training pulls their vectors together.

	ONE-HOT	DENSE EMBEDDING
Length	vocabulary size (~50,000)	small and fixed (50-300)
Values	zeros and one 1	real numbers
Created by	indexing	learned from data
Captures Meaning	No	Yes
Similar words close	no - all equidistant	Yes

2. Word2Vec, GloVe and FastText

⇒ All three rest on the distributional hypothesis:

> "You shall know a word by the company it keeps."
>
> - J.R. Firth, 1957

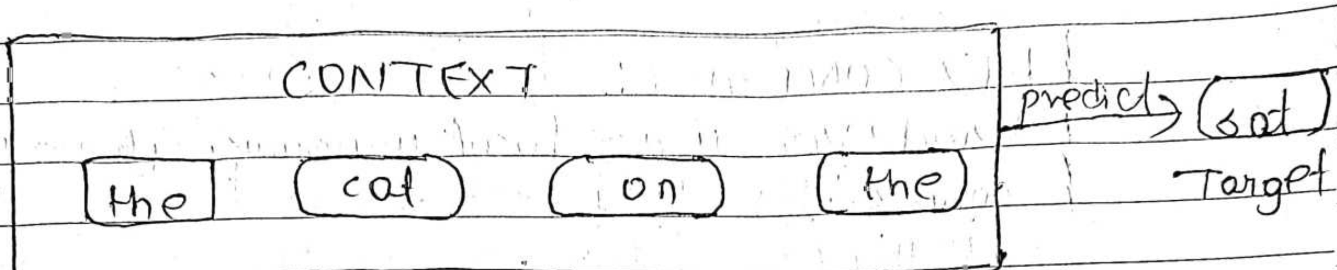
Words used in similar contexts have similar meanings.

⇒ Word2Vec (2013)

Slides a window over the text and

trains a small neural network. For the sentence, the cat sat on the mat with center word sat, the context is [the, cat, on, the]. There are two ways round:

* CBOW - context predicts the word



* SKIP-GRAM - word predicts the context

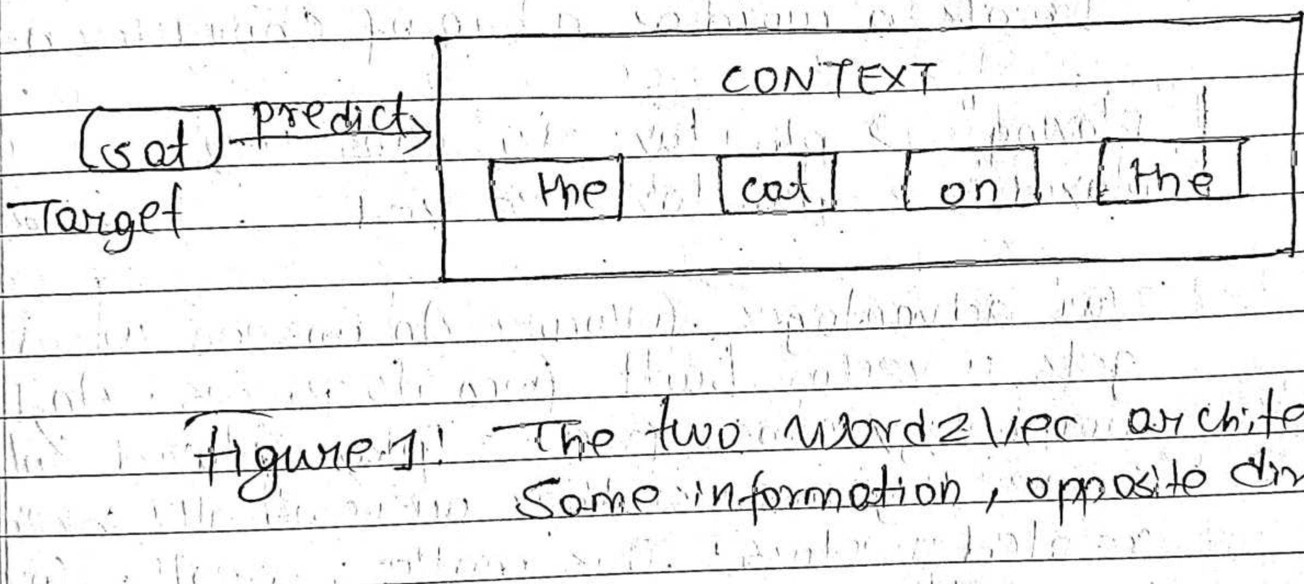


Figure 1: The two word2vec architectures. Same information, opposite direction.

	CBOW	SKIP-GRAM
Direction	context → word	word → context
Speed	faster	slower
Rare words	weaker	better
Small corpus	weaker	better

Date
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* GloVe (2014) - Global Vectors

Word2Vec sees only one window at a time. GloVe instead first counts how often every pair of words co-occurs across the whole corpus, then factorises that count matrix.

> KEY CONTRAST:
> Word2Vec uses local windows; GloVe uses global counts.

* FastText (2016)

Treats a word as a bag of character n-grams:

"playing" → pla lay ayi yin ing
"played" → pla lay aye yed ← shares pla lay

Two advantages follow. An unseen word still gets a vector built from its pieces. And morphology comes free — play / played / playing share n-grams, so they automatically receive related vectors. This matters greatly for morphologically rich languages such as Nepali.

MODEL	CORE IDEA	STRENGTH
Word2Vec	predict within a local window	fast, simple
GloVe	factorise global co-occurrence count	whole-corpus statistics
FastText	word = sum of character n-grams	unseen words, morphology.

3. Semantic Similarity

Closeness of meaning is measured by cosine similarity - the angle between two vectors, ignoring their length:

$$\cos(A,B) = \frac{A \cdot B}{|A| \times |B|}$$

-1 opposite
0 unrelated
+1 identical

* Why the angle, and not the distance?

Frequent words develop longer vectors. Plain straight-line distance would call a common word "far" from a rare one purely because of frequency. The angle discards length and keeps only direction and meaning lives in the direction.

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* Measured on GloVe (100 dimensions, 400,000 words)

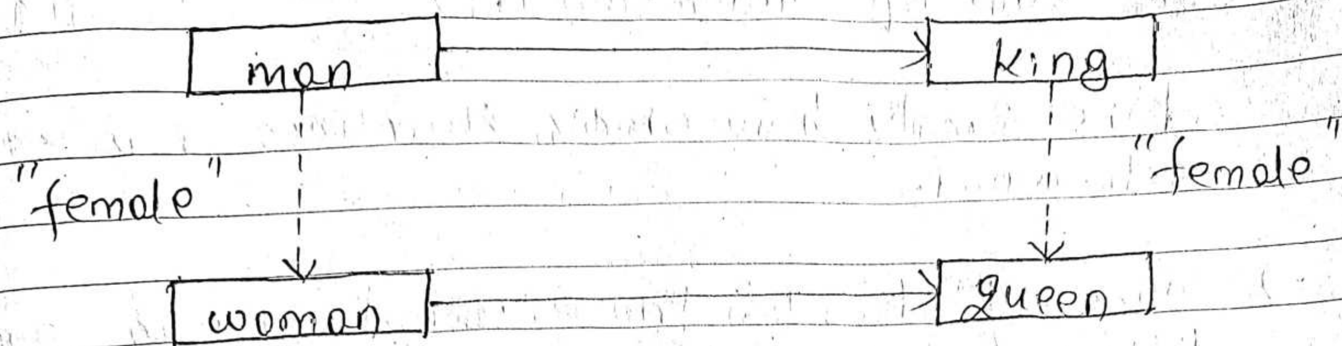
PAIR	COS	
cat - dog	0.88	same contexts
good - bad	0.77	see warning below
King - queen	0.75	both royalty
cat - rocket	0.19	unrelated

WARNING - AN IMPORTANT LIMITATION

good - bad scores 0.77, nearly as high as King - queen, even though they are opposites. Cosine similarity really measures "appears in similar contexts", not "means the same thing". Both words fit the sentence "this film was very ----". So static embeddings cannot separate antonyms from synonyms.

* Vector Arithmetic

Relationships between words appear as consistent directions in the space.



both horizontal arrows have the same dimension and length

Figure 2: Relationships as directions: king − man + woman ≃ queen.

ARITHMETIC	TOP RESULT	SCORE
king − man + woman	queen	0.77
paris − france + nepal	kathmandu	0.81

The vectors were never told what a capital city is. They learned the relationship country → capital from usage alone, and it transfers correctly to Nepal.

4. The Role of Dimensionality

d is simply how many numbers represent each word.

→ Too small (d=5) : there is not enough room, so different meanings collide — the model underfits.
→ Too large (d = 1000) : each dimension sees too little data, so the model memorises noise instead of meaning — it overfits — and costs more money and time.

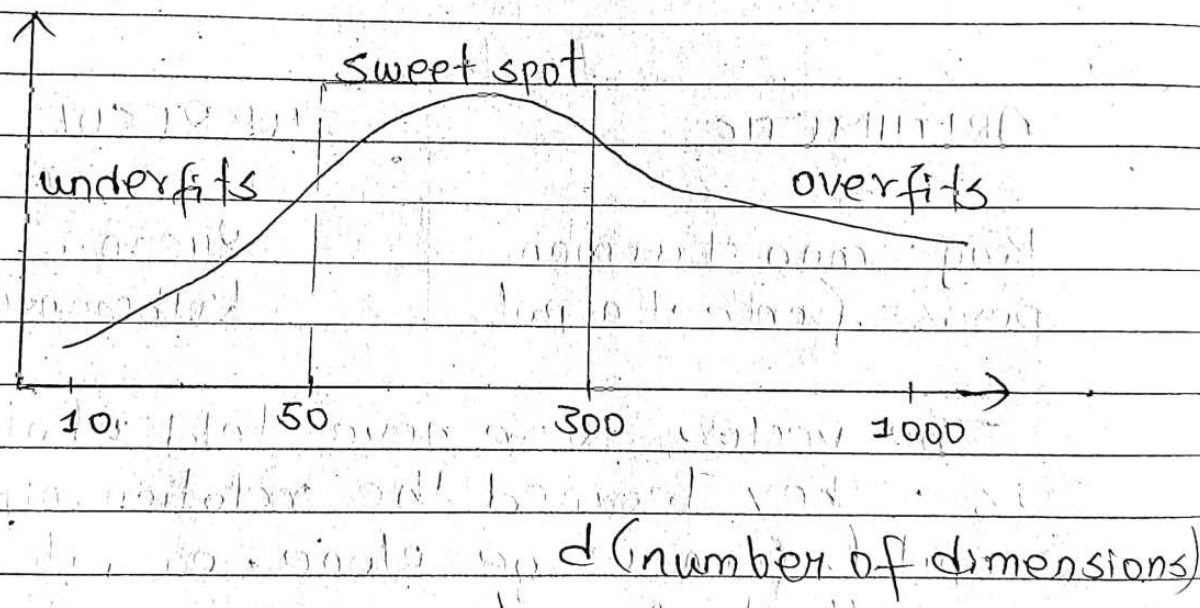


Figure 3: Quality against dimensionality. Both extremes are worse than the middle.

D	EFFECT
<20	underfits — different meanings collide
50-300	the usual sweet spot
>1000	overfits, slower, diminishing returns

IMPORTANT

Individual dimensions are not interpretable. Dimension 42 does not mean "royalty". Meaning is spread across the whole vector and appear as directions through the space, not along single axes. This is why embeddings work well but are hard to explain.